

PERTEMUAN KE – 7 SQL

A. TUJUAN

- Memahami perintah perintah dasar SQL untuk CRUD

B. TEORI SINGKAT

Database-Level

```
DROP DATABASE databaseName
DROP DATABASE IF EXISTS databaseName
CREATE DATABASE databaseName
CREATE DATABASE IF NOT EXISTS databaseName
SHOW DATABASES
USE databaseName
SELECT DATABASE()
SHOW CREATE DATABASE databaseName
```

Table-Level

```
DROP TABLE [IF EXISTS] tableName, ...
CREATE TABLE [IF NOT EXISTS] tableName (
    columnName columnType columnAttribute, ...
    PRIMARY KEY(columnName),
    FOREIGN KEY (columnNmae) REFERENCES tableName (columnNmae)
)
SHOW TABLES
DESCRIBE|DESC tableName
ALTER TABLE tableName ...
ALTER TABLE tableName ADD columnDefinition
ALTER TABLE tableName DROP columnName
ALTER TABLE tableName ADD FOREIGN KEY (columnNmae) REFERENCES tableName
(columnNmae)
ALTER TABLE tableName DROP FOREIGN KEY constraintName
SHOW CREATE TABLE tableName
```

Row-Level

```
INSERT INTO tableName
    VALUES (column1Value, column2Value,...)
INSERT INTO tableName
    VALUES (column1Value, column2Value,...), ...
INSERT INTO tableName (column1Name, ..., columnNName)
    VALUES (column1Value, ..., columnNValue)
DELETE FROM tableName WHERE criteria
UPDATE tableName SET columnName = expr, ... WHERE criteria
SELECT * | column1Name AS alias1, ..., columnNName AS aliasN
    FROM tableName
    WHERE criteria
    GROUP BY columnName
    ORDER BY columnName ASC|DESC, ...
    HAVING groupConstraints
    LIMIT count | offset count
```

C. PRAKTIK

Praktik 1

Membuat dan menghapus database

```
>mysql -u root -p
password .....
```

```

mysql> SHOW DATABASES;
mysql> CREATE DATABASE storedb;

mysql> DROP DATABASE storedb;

mysql> CREATE DATABASE IF NOT EXISTS storedb;

mysql> DROP DATABASE IF EXISTS storedb;

mysql> CREATE DATABASE IF NOT EXISTS storedb;

mysql> SHOW CREATE DATABASE storedb \G

mysql> DROP DATABASE IF EXISTS storedb;

mysql> SHOW DATABASES;

mysql> QUIT;

```

Praktik 2 **Membuat user database**

masuk sebagai root

```

>mysql -u root -p
password .....

```

```

mysql> SHOW DATABASES;

```

```

mysql> CREATE DATABASE `storedb` ;

```

```

mysql> use mysql;

```

```

mysql> INSERT INTO `user` ( `Host` , `User` , `Password` )
VALUES (
    '%', 'store', PASSWORD( 'store' )
);

```

```

mysql> INSERT INTO `db`
VALUES (
    '%', 'storedb', 'store', 'Y', 'Y', 'Y', 'Y', 'Y', 'Y', 'Y', 'Y', 'Y', 'Y', 'Y',
    'Y', 'Y', 'Y', 'Y', 'Y', 'Y', 'Y', 'Y', 'Y'
);

```

restart service mysql

masuk sebagai store

```

>mysql -u store -p
password .....

```

```

mysql> SHOW DATABASES;

```

Harusnya hanya terlihat database storedb

Praktik 3

Membuat tabel products

Database: storedb Table: products				
id INT	productCode CHAR(3)	name VARCHAR(30)	quantity INT	price DECIMAL(10,2)
1001	PEN	Pen Red	5000	1.23
1002	PEN	Pen Blue	8000	1.25
1003	PEN	Pen Black	2000	1.25
1004	PEC	Pencil 2B	10000	0.48
1005	PEC	Pencil 2H	8000	0.49

```
mysql> USE storedb;
```

```
mysql> SELECT DATABASE();
```

```
mysql> SHOW TABLES;
```

```
mysql> CREATE TABLE IF NOT EXISTS products (  
    id      INT UNSIGNED NOT NULL AUTO_INCREMENT,  
    productCode CHAR(3) NOT NULL DEFAULT '',  
    name     VARCHAR(30) NOT NULL DEFAULT '',  
    quantity INT UNSIGNED NOT NULL DEFAULT 0,  
    price    DECIMAL(7,2) NOT NULL DEFAULT 99999.99,  
    PRIMARY KEY (id)  
);
```

```
mysql> SHOW TABLES;
```

```
mysql> DESCRIBE products;
```

```
mysql> SHOW CREATE TABLE products \G
```

Praktik 4

Inserting Rows - INSERT INTO

```
mysql> INSERT INTO products VALUES (1001, 'PEN', 'Pen Red', 5000, 1.23);
```

```
mysql> INSERT INTO products VALUES  
    (NULL, 'PEN', 'Pen Blue', 8000, 1.25),  
    (NULL, 'PEN', 'Pen Black', 2000, 1.25);
```

```
mysql> INSERT INTO products (productCode, name, quantity, price) VALUES  
    ('PEC', 'Pencil 2B', 10000, 0.48),  
    ('PEC', 'Pencil 2H', 8000, 0.49);
```

```
mysql> INSERT INTO products (productCode, name) VALUES ('PEC', 'Pencil  
HB');
```

```
mysql> INSERT INTO products values (NULL, NULL, NULL, NULL, NULL);
```

```
mysql> SELECT * FROM products;
```

```
mysql> DELETE FROM products WHERE id = 1006;
```

Praktik 5

Querying the Database - SELECT

```
mysql> SELECT name, price FROM products;
```

```
mysql> SELECT * FROM products;
```

SELECT without Table

```
mysql> SELECT 1+1;
```

```
mysql> SELECT NOW();
```

```
mysql> SELECT 1+1, NOW();
```

Comparison Operators

Untuk numbers (INT, DECIMAL, FLOAT): comparison operators:

- '=' (equal to),
- '<>' or '!=' (not equal to),
- '>' (greater than),
- '<' (less than),
- '>=' (greater than or equal to),
- '<=' (less than or equal to)

```
mysql> SELECT name, price FROM products WHERE price < 1.0;
```

```
mysql> SELECT name, quantity FROM products WHERE quantity <= 2000;
```

```
mysql> SELECT name, price FROM products WHERE productCode = 'PEN';
```

String Pattern Matching - LIKE and NOT LIKE

Untuk strings, *pattern matching*

- 'abc%' matches strings beginning with 'abc';
- '%xyz' matches strings ending with 'xyz';
- '%aaa%' matches strings containing 'aaa';
- '____' matches strings containing exactly three characters; and
- 'a_b%' matches strings beginning with 'a', followed by any single character, followed by 'b', followed by zero or more characters.

```
mysql> SELECT name, price FROM products WHERE name LIKE 'PENCIL%';
```

```
mysql> SELECT name, price FROM products WHERE name LIKE 'P__ %';
```

Arithmetic Operators

Operator	Description
+	Addition
-	Subtraction
*	Multiplication
/	Division
DIV	Integer Division
%	Modulus (Remainder)

Logical Operators - AND, OR, NOT, XOR

```
mysql> SELECT * FROM products WHERE quantity >= 5000 AND name LIKE 'Pen %';
```

```
mysql> SELECT * FROM products WHERE quantity >= 5000 AND price < 1.24 AND  
name LIKE 'Pen %';
```

```
mysql> SELECT * FROM products WHERE NOT (quantity >= 5000 AND name LIKE  
'Pen %');
```

IN, NOT IN

```
mysql> SELECT * FROM products WHERE name IN ('Pen Red', 'Pen Black');
```

BETWEEN, NOT BETWEEN

```
mysql> SELECT * FROM products  
WHERE (price BETWEEN 1.0 AND 2.0) AND (quantity BETWEEN 1000 AND  
2000);
```

IS NULL, IS NOT NULL

```
mysql> SELECT * FROM products WHERE productCode IS NULL;
```

ORDER BY Clause

```
mysql> SELECT * FROM products WHERE name LIKE 'Pen %' ORDER BY price DESC;
```

```
mysql> SELECT * FROM products WHERE name LIKE 'Pen %' ORDER BY price DESC,  
quantity;
```

```
mysql> SELECT * FROM products ORDER BY RAND();
```

LIMIT Clause

```
mysql> SELECT * FROM products ORDER BY price LIMIT 2;
```

```
mysql> SELECT * FROM products ORDER BY price LIMIT 2, 1;
```

AS - Alias

```
mysql> SELECT id AS ID, productCode AS Code,  
name AS Description, price AS `Unit Price`  
FROM products  
ORDER BY ID;
```

Function CONCAT()

```
mysql> SELECT CONCAT(productCode, ' - ', name) AS `Product Description`,  
price FROM products;
```

DISTINCT

```
mysql> SELECT price FROM products;
```

```
mysql> SELECT DISTINCT price AS `Distinct Price` FROM products;
```

```
mysql> SELECT DISTINCT price, name FROM products;
```

GROUP BY Clause

```
mysql> SELECT * FROM products ORDER BY productCode, id;
```

```
mysql> SELECT * FROM products GROUP BY productCode;
```

GROUP BY Aggregate Functions: COUNT, MAX, MIN, AVG, SUM, STD, GROUP_CONCAT

```
mysql> SELECT COUNT(*) AS `Count` FROM products;
```

```
mysql> SELECT productCode, COUNT(*) FROM products GROUP BY productCode;
```

```
mysql> SELECT productCode, COUNT(*) AS count
        FROM products
        GROUP BY productCode
        ORDER BY count DESC;
```

```
mysql> SELECT MAX(price), MIN(price), AVG(price), STD(price), SUM(quantity)
        FROM products;
```

```
mysql> SELECT productCode, MAX(price) AS `Highest Price`, MIN(price) AS
        `Lowest Price`
        FROM products
        GROUP BY productCode;
```

```
mysql> SELECT productCode, MAX(price), MIN(price),
        CAST(AVG(price) AS DECIMAL(7,2)) AS `Average`,
        CAST(STD(price) AS DECIMAL(7,2)) AS `Std Dev`,
        SUM(quantity)
        FROM products
        GROUP BY productCode;
```

HAVING clause

```
mysql> SELECT
        productCode AS `Product Code`,
        COUNT(*) AS `Count`,
        CAST(AVG(price) AS DECIMAL(7,2)) AS `Average`
        FROM products
        GROUP BY productCode
        HAVING Count >=3;
```

WITH ROLLUP

```
mysql> SELECT
        productCode,
        MAX(price),
        MIN(price),
        CAST(AVG(price) AS DECIMAL(7,2)) AS `Average`,
        SUM(quantity)
        FROM products
        GROUP BY productCode
        WITH ROLLUP;
```

Praktik 6

Modifying Data - UPDATE

```
mysql> UPDATE products SET price = price * 1.1;

mysql> SELECT * FROM products;

mysql> UPDATE products SET quantity = quantity - 100 WHERE name = 'Pen
Red';

mysql> SELECT * FROM products WHERE name = 'Pen Red';

mysql> UPDATE products SET quantity = quantity + 50, price = 1.23 WHERE
name = 'Pen Red';

mysql> SELECT * FROM products WHERE name = 'Pen Red';
```

Praktik 7

Deleting Rows - DELETE FROM

```
mysql> DELETE FROM products WHERE name LIKE 'Pencil%';

mysql> SELECT * FROM products;

mysql> DELETE FROM products;

mysql> SELECT * FROM products;
```

Praktik 8

Loading/Exporting Data from/to a Text File

LOAD DATA LOCAL INFILE ... INTO TABLE ...

```
mysql> LOAD DATA LOCAL INFILE 'd:/myProject/products_in.csv' INTO TABLE
products
        COLUMNS TERMINATED BY ','
        LINES TERMINATED BY '\r\n';
```

```
mysql> SELECT * FROM products;
```

mysqlimport Utility Program

```
> mysqlimport -u username -p --local databaseName tableName.tsv
> cd path-to-mysql-bin
> mysqlimport -u root -p --local storedb d:/myProject/products.tsv
```

SELECT ... INTO OUTFILE ...

```
mysql> SELECT * FROM products INTO OUTFILE 'd:/myProject/products_out.csv'
        COLUMNS TERMINATED BY ','
        LINES TERMINATED BY '\r\n';
```

Praktik 9
Multiple Tables
One-To-Many Relationship

Database: storedb Table: suppliers		
supplierID INT	name VARCHAR(3)	phone CHAR(8)
501	ABC Traders	88881111
502	XYZ Company	88882222
503	QQ Corp	88883333

Database: storedb Table: products					
id INT	ProductCode CHAR(3)	Name VARCHAR(30)	Quantity INT	Price DECIMAL(10,2)	SupplierID INT (Foreign Key)
2001	PEC	Pencil 3B	500	0.52	501
2002	PEC	Pencil 4B	200	0.62	501
2003	PEC	Pencil 5B	100	0.73	501
2004	PEC	Pencil 6B	500	0.47	502

```
mysql> USE storedb;
```

```
mysql> DROP TABLE IF EXISTS suppliers;
```

```
mysql> CREATE TABLE suppliers (
    supplierID INT UNSIGNED NOT NULL AUTO_INCREMENT,
    name       VARCHAR(30)  NOT NULL DEFAULT '',
    phone      CHAR(8)      NOT NULL DEFAULT '',
    PRIMARY KEY (supplierID)
);
```

```
mysql> DESCRIBE suppliers;
```

```
mysql> INSERT INTO suppliers VALUE
    (501, 'ABC Traders', '88881111'),
    (502, 'XYZ Company', '88882222'),
    (503, 'QQ Corp', '88883333');
```

```
mysql> SELECT * FROM suppliers;
```

ALTER TABLE

```
mysql> ALTER TABLE products
    ADD COLUMN supplierID INT UNSIGNED NOT NULL;
```

```
mysql> DESCRIBE products;
```

```
mysql> UPDATE products SET supplierID = 501;
```

```
mysql> ALTER TABLE products
    ADD FOREIGN KEY (supplierID) REFERENCES suppliers (supplierID);
```

```
mysql> DESCRIBE products;
```



```
mysql> UPDATE products SET supplierID = 502 WHERE id = 2004;
```

```
mysql> SELECT * FROM products;
```

SELECT with JOIN

```
mysql> SELECT products.name, price, suppliers.name  
FROM products  
JOIN suppliers ON products.supplierID = suppliers.supplierID  
WHERE price < 0.6;
```

```
mysql> SELECT products.name, price, suppliers.name  
FROM products, suppliers  
WHERE products.supplierID = suppliers.supplierID  
AND price < 0.6;
```

```
mysql> SELECT products.name AS `Product Name`, price, suppliers.name AS  
`Supplier Name`  
FROM products  
JOIN suppliers ON products.supplierID = suppliers.supplierID  
WHERE price < 0.6;
```

```
mysql> SELECT p.name AS `Product Name`, p.price, s.name AS `Supplier Name`  
FROM products AS p  
JOIN suppliers AS s ON p.supplierID = s.supplierID  
WHERE p.price < 0.6;
```

D. LATIHAN

- Latihan diberikan oleh dosen pengampu pada saat praktikum.
- Dikerjakan di laboratorium pada jam praktikum.

E. TUGAS

- Tugas diberikan oleh dosen pengampu pada akhir praktikum.
- Dikerjakan di rumah dan dilampirkan pada laporan.